Dynamic Benchmarking of LLMs: Moving Beyond Static Evaluation

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Abstract

The rapid advancement of Large Language Models (LLMs) has outpaced the ability of traditional static benchmarks to evaluate them effectively. As models are trained on increasingly vast internet corpora, static benchmarks are severely contaminated by test set leakage, making it difficult to determine whether high scores reflect genuine reasoning or mere data regurgitation. This study investigates how model performance and rankings shift when moving from traditional static evaluations to dynamic and metalinguistic evaluations. Three frontier models released in 2026—GPT-5.4, Claude Sonnet 4.6, and MiniMax M2.7—were tested across three dimensions: a static knowledge benchmark (MMLU), a dynamic, contamination-limited benchmark (LiveBench), and an expert-level metalinguistic dataset. The results reveal a severe performance drop between static and dynamic testing environments. While all three models achieved near-tied scores of ~88% on the static MMLU, their performance plummeted on LiveBench, with Claude Sonnet 4.6 leading at 40.0%, GPT-5.4 at 37.2%, and MiniMax M2.7 dropping drastically to 30.0%. Furthermore, metalinguistic evaluations revealed distinct capability footprints, with GPT-5.4 demonstrating the strongest theoretical linguistic analysis, achieving 66.0% accuracy. These findings strongly suggest that static benchmark scores are artificially inflated by test set leakage and that future AI evaluations in computational linguistics must use frequently updated, expert-level tasks to gauge true generalization and reasoning capabilities accurately.

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1. Introduction

Large Language Models (LLMs) have become foundational to modern artificial intelligence technologies, yet reliably evaluating their true language comprehension and reasoning capabilities remains a critical challenge (Liang et al., 2022). Traditionally, LLM performance is assessed using static benchmarks composed of fixed, multiple-choice questions. A prominent example is the Measuring Massive Multitask Language Understanding (MMLU) benchmark, which was designed to evaluate multitask accuracy across 57 academic and professional subjects, including elementary mathematics, United States history, computer science, and law (Hendrycks et al., 2021). Because the MMLU relies entirely on a multiple-choice format in which models select from four static options, aggregate performance is measured simply as the total percentage of correct selections (Hendrycks et al., 2021).

2. Background

A fundamental obstacle in current evaluation methodologies is the inherent vulnerability of static tests to data contamination (Li et al., 2024). Because contemporary models ingest massive portions of the internet during pretraining, they face a severe risk of test set leakage. Researchers note that test set contamination—wherein test data from a benchmark inadvertently becomes part of a newer model’s training corpus—is a well-documented obstacle for fair evaluation that can quickly render static benchmarks obsolete (White et al., 2024). Consequently, when an AI system is evaluated on a contaminated benchmark, exceptionally high scores often reflect the model's statistical familiarity with memorized test content rather than genuine, independent reasoning skills. This phenomenon compromises the validity of traditional static evaluations, as it artificially inflates performance metrics and makes newer models appear far more capable than their actual real-world problem-solving abilities would suggest (White et al., 2024). To overcome this illusion of competence, evaluation frameworks must pivot toward continuously updated test sets and entirely novel, expert-level problems that definitively separate rote memorization from active, abstract reasoning (Beguš et al., 2025; White et al., 2024).

2.1 The Shift Toward Dynamic Benchmarks

To effectively circumvent data contamination, researchers have developed "dynamic" benchmarking frameworks that refresh their test sets frequently (White et al., 2024). Dynamic evaluations, such as LiveBench, combat data contamination by continuously introducing novel questions derived from recent information sources and scoring answers automatically according to objective ground-truth values (White et al., 2024). Because the test questions are constantly changing, AI models cannot memorize the answers in advance during pretraining. This forces the models to engage with entirely new problems, providing a much clearer and more honest assessment of their genuine reasoning capabilities under real-world conditions (White et al., 2024).

However, maintaining dynamic evaluations presents significant logistical challenges. Creating tests that continuously update is resource-intensive, and even newer task-based benchmarks often contain underlying components that remain relatively static (Jimenez et al., 2024). Consequently, without rigorous and continuous updates, even dynamic tests can eventually face the same data contamination issues over time. Furthermore, despite the development of these dynamic metrics, it remains surprisingly rare in the current literature to comprehensively compare new dynamic evaluations side-by-side with older static tests under identical, strictly controlled experimental conditions (White et al., 2024).

Additionally, evaluating complex, open-ended responses—such as formal metalinguistic analyses—introduces a scoring dilemma, as these tasks lack simple objective ground truths for programmatic grading. To solve this, researchers are increasingly utilizing the "LLM-as-a-judge" paradigm, where a highly capable language model uses pre-defined criteria to act as the evaluator (Zheng et al., 2024). While frameworks like LiveBench explicitly avoid this method by relying exclusively on objective, verifiable ground-truth values (White et al., 2024), the LLM-as-a-judge approach provides a highly scalable and relatively reliable way to score open-ended benchmarks and complex linguistic tasks that would otherwise require prohibitive amounts of expert human labor (Li et al., 2024).

2.2 Evaluating Metalinguistic Abilities in LLMs

Furthermore, within the specific domain of computational linguistics, evaluating a model's true capability requires moving beyond general knowledge trivia to assess formal metalinguistic abilities (Beguš et al., 2025). Metalinguistic ability is defined as the capacity to actively analyze language itself and generate formal, theoretical analyses of linguistic phenomena, rather than merely engaging in behavioral language use (Beguš et al., 2025). Because most standard artificial intelligence benchmarks focus on broad trivia or basic mathematical reasoning (Hendrycks et al., 2021), they fail to adequately evaluate how well a model performs on expert-level linguistic abstraction.

Comprehensive formal linguistic evaluation encompasses several highly specialized domains: syntactic analysis tests whether a model can accurately comprehend the structural rules of language by generating hierarchical syntactic trees; semantic analysis evaluates how a model handles meaning, specifically its ability to formally resolve structural ambiguity across different sentence parses rather than interpreting words merely at face value; metaphorical analysis assesses a model's capacity to interpret nuanced figurative language in context; and phonological analysis examines how well a text-based system understands underlying sound architectures, such as identifying syllable structures, generalizing sound patterns, and applying formal phonological rules (Beguš et al., 2025).

To address these critical evaluation gaps, this study asks the following research question: Compared with traditional static evaluation benchmarks like the MMLU (Hendrycks et al., 2021), how do dynamic evaluations like LiveBench (White et al., 2024) and formal metalinguistic evaluations (Beguš et al., 2025) alter the performance rankings and assessed reasoning capabilities of contemporary LLMs? I hypothesize that while frontier models will exhibit high performance on static tests, the massive performance gaps and inflated scores seen on static leaderboards will shrink under dynamic and metalinguistic conditions. In these uncontaminated environments, models can no longer rely on memorized training data and must instead demonstrate genuine, generalized reasoning (Beguš et al., 2025; White et al., 2024)

3. Methods

In this study, we evaluate three state-of-the-art frontier language models, all released in early 2026, to capture the most current landscape of artificial intelligence capabilities while optimizing for budget and inference speed. The selected models are OpenAI's GPT-5.4, Anthropic's Claude Sonnet 4.6, and MiniMax's M2.7 (Anthropic, 2026; MiniMax, 2026; OpenAI, 2026). These specific systems were chosen to provide a balanced comparative analysis across proprietary architectures and highly efficient, cost-effective alternatives.

3.1 Selected Models and Setup

GPT-5.4, released on March 5, 2026, serves as the flagship, general-purpose baseline, bringing together recent advances in reasoning, coding, and agentic computer-use capabilities into a single model (OpenAI, 2026). Claude Sonnet 4.6 is included as a highly capable model that substantially improves upon its predecessor, approaching or matching the capability levels of Anthropic's frontier Opus 4.6 model while remaining highly cost-effective (Anthropic, 2026). Finally, MiniMax M2.7 is included as an extremely efficient frontier alternative; remarkably, it acts as the first model to deeply participate in its own recursive self-evolution during the training process (MiniMax, 2026). To ensure a fair comparison, all models will be tested under uniform conditions. Furthermore, following recent methodological precedents in formal linguistic evaluation, the models' temperature parameters will remain unchanged, as "within the range of 0.0–1.0, changes in temperature do not affect the LLMs’ problem-solving ability" (Beguš et al., 2025).

3.2 Evaluation Benchmarks

To comprehensively assess these models, they will be evaluated across three distinct benchmarking frameworks, each selected to measure a different dimension of reasoning and language comprehension. First, the Measuring Massive Multitask Language Understanding (MMLU) dataset will be utilized as the traditional, industry-recognized baseline for static evaluation (Hendrycks et al., 2021). Second, LiveBench will be deployed as a rigorous, continuously updated dynamic benchmark capable of testing the models' general reasoning and knowledge-based capabilities without the confounding variable of test set contamination (White et al., 2024). Finally, the models' formal linguistic understanding and analytical reasoning will be evaluated using the expert-level metalinguistic dataset introduced by Beguš et al., which actively tests capabilities across multiple theoretical linguistic domains (Beguš et al., 2025).

4. Results

The primary objective of this study was to determine how dynamic evaluations (White et al., 2025) and formal metalinguistic assessments (Beguš et al., 2025) alter the performance rankings and assessed reasoning capabilities of contemporary large language models compared to traditional static benchmarks like the MMLU (Hendrycks et al., 2021). The evaluation yielded a comprehensive dataset across these three distinct testing frameworks, revealing significant performance shifts when models were deprived of contaminated training data.

4.1 Baseline Performance on Static Evaluations

Under traditional static testing conditions, all three frontier models exhibited exceptionally high accuracy, consistent with performance metrics frequently touted on industry leaderboards (Hendrycks et al., 2021). As shown in Figure 1, on the Measuring Massive Multitask Language Understanding (MMLU) benchmark, Anthropic's Claude Sonnet 4.6 established a dominant baseline with an accuracy of 95.8% (Anthropic, 2026). MiniMax M2.7 followed with 88.5% (MiniMax, 2026), and OpenAI's GPT-5.4 recorded an accuracy of 87.5% (OpenAI, 2026).

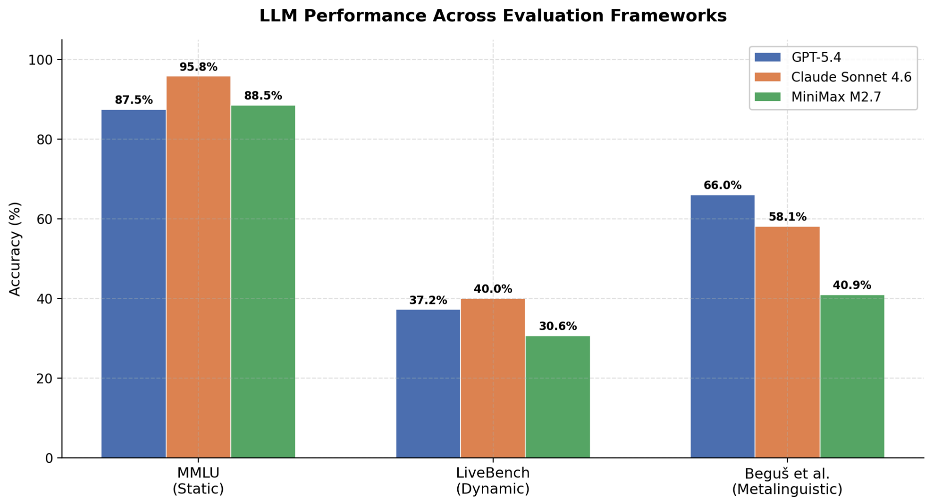

Figure 1: LLM Performance Across Evaluation Frameworks

These high static scores initially suggest robust, generalized intelligence across all three systems. However, examining the distribution of accuracy across all benchmarks reveals that this static baseline artificially inflates the models' genuine reasoning capabilities.

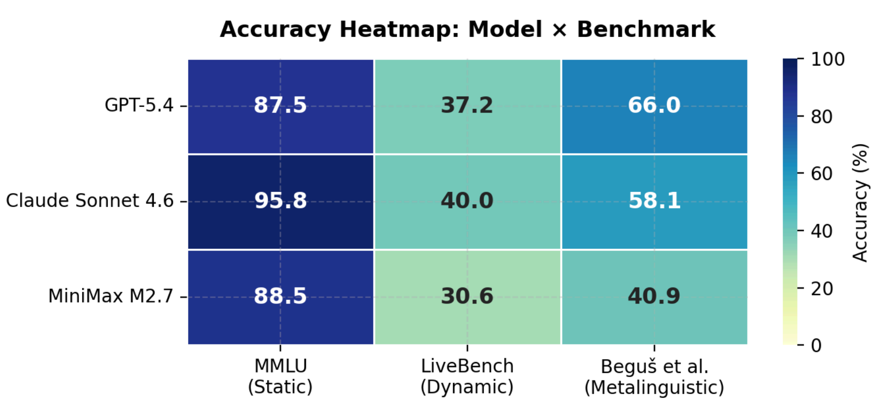

Figure 2: Accuracy Heatmap: Model x benchmark

4.2 The Data Contamination Effect on Dynamic Benchmarks

When transitioning from the static MMLU to the continuously updated, dynamic LiveBench framework, all three models experienced a severe, immediate performance degradation (White et al., 2025). Claude Sonnet 4.6's accuracy plummeted from 95.8% to 40.0%, representing a massive 55.8% drop. Similarly, GPT-5.4 fell to 37.2% (a 50.3% drop), and MiniMax M2.7 dropped to 30.6% (a 57.9% drop).

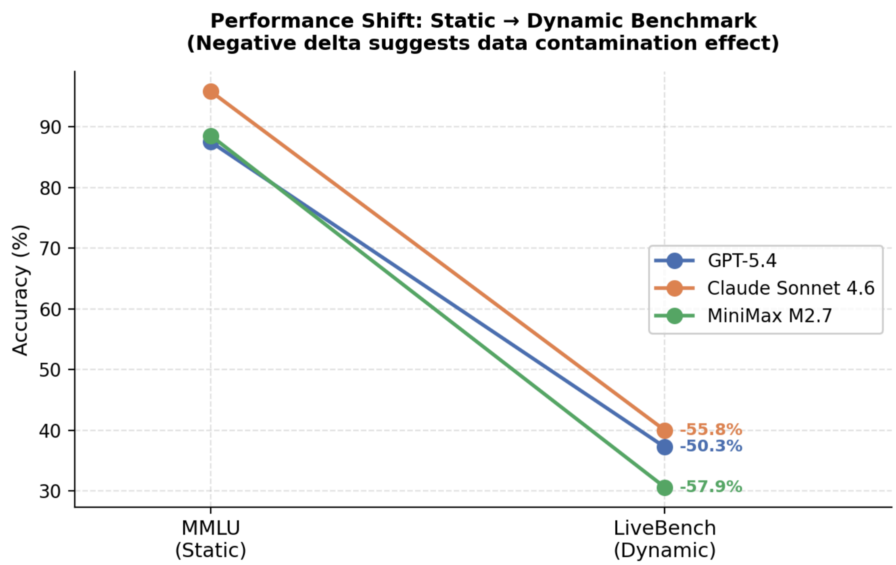

Figure 3: Performance Shift from Static to Dynamic Benchmark

This precipitous decline provides strong empirical evidence for the data contamination hypothesis. Because LiveBench continuously introduces novel questions from recent sources, models cannot rely on test set leakage from their pretraining corpora (White et al., 2025). The resulting negative performance delta across all models confirms that traditional static leaderboards heavily overstate independent reasoning skills, masking the extent to which models rely on statistical familiarity with memorized test content.

The evaluation of formal metalinguistic abilities further disrupted the initial performance rankings established by the static MMLU. While Claude Sonnet 4.6 led the static benchmark, GPT-5.4 demonstrated superior capability in abstract linguistic reasoning. On the metalinguistic tasks introduced by Beguš et al. (2025)—which required the models to generate formal syntactic trees, resolve semantic ambiguity, and perform phonological analysis—GPT-5.4 achieved the highest accuracy at 66.0%. Claude Sonnet 4.6 fell to second place with 58.1%, and MiniMax M2.7 struggled significantly with the expert-level abstractions, scoring only 40.9%.

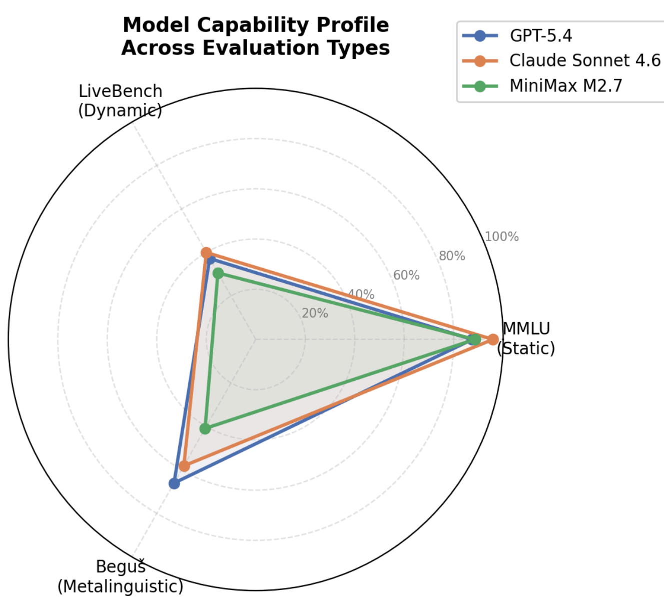

Figure 4: Model Capability Profile Across Evaluation Types

Ultimately, the empirical results confirm the study's central hypothesis: the massive performance gaps and inflated scores seen on static leaderboards shrink dramatically under dynamic and metalinguistic conditions (Beguš et al., 2025; White et al., 2025). Furthermore, exposing these models to entirely novel, uncontaminated environments actively alters their performance rankings. As shown in the model capability profile (Figure 4), a model's proficiency in recalling static trivia (MMLU) does not reliably translate into superior performance on active, generalized reasoning tasks (LiveBench and metalinguistic analysis).

5. Discussions

The findings of this study provide compelling evidence* that traditional, static evaluation metrics fundamentally misrepresent the genuine reasoning capabilities of contemporary large language models. By comparing model performance across static, dynamic, and expert-level metalinguistic frameworks, this research highlights the severe impact of data contamination and underscores the necessity of continuous, domain-specific evaluation.

5.1 The Illusion of Static Competence

The most striking observation from the dataset is the precipitous performance degradation all models experienced when transitioning from the static MMLU benchmark (Hendrycks et al., 2021) to the dynamic LiveBench framework (White et al., 2025). While Anthropic's Claude Sonnet 4.6 achieved a near-perfect 95.8% on the MMLU, its accuracy collapsed to 40.0% on LiveBench. Similar performance drops of over 50% were observed for both GPT-5.4 (which fell to 37.2%) and MiniMax M2.7 (which fell to 30.6%).

This massive negative delta definitively confirms the hypothesis that exceptionally high scores on established static benchmarks are largely an artifact of test set leakage (White et al., 2025). Because frontier models ingest vast portions of the internet during pretraining, static multiple-choice questions primarily measure statistical familiarity and memorization rather than active problem-solving (Hendrycks et al., 2021). Dynamic evaluations like LiveBench successfully mitigate this illusion of competence by continuously refreshing their prompts, thereby providing a far more realistic assessment of a model's true, uncontaminated reasoning skills under real-world conditions (White et al., 2025).

5.2 Expert-Level Metalinguistic Capabilities

Beyond the static-dynamic divide, the evaluation of formal metalinguistic abilities revealed significant disparities in how models process deep linguistic abstractions. Standard benchmarks rarely test expert-level tasks like building hierarchical syntactic trees, formalizing semantic ambiguity, or applying generalized phonological rules to novel words (Beguš et al., 2025). These metalinguistic tasks are highly complex because they require a model to explicitly reason about the structure of language itself, rather than merely generating highly probable conversational tokens based on behavioral language use (Beguš et al., 2025).

In this uncontaminated, expert-level domain, the performance rankings inverted from the static baseline. GPT-5.4 demonstrated superior metacognitive linguistic abilities, achieving the highest accuracy at 66.0% (OpenAI, 2026). In contrast, Claude Sonnet 4.6 fell to 58.1% (Anthropic, 2026), and the highly efficient MiniMax M2.7 struggled significantly with the theoretical abstractions, scoring only 40.9% (MiniMax, 2026).

This shift in rankings indicates that a model's capacity to recall general knowledge trivia or execute standard programming tasks does not inherently translate to complex theoretical linguistics (Beguš et al., 2025). The ability to apply formal linguistic frameworks to novel, unobserved language data remains a distinct and highly challenging cognitive task for artificial intelligence (Beguš et al., 2025). While frontier models like GPT-5.4 show promising early indicators of generalized metalinguistic reasoning, the overall scores—all remaining well below 70%—suggest that the field of computational linguistics still presents a formidable frontier for AI development.

Ultimately, to accurately gauge the progression of artificial intelligence, researchers must continue to pivot away from static trivia and toward dynamic, expert-calibrated evaluations that definitively separate rote memorization from genuine, abstract reasoning.

5.3 Granular Analysis of Metalinguistic Performance

A closer examination of the models' performance across the specific subtasks within the metalinguistic dataset reveals a stark contrast between surface-level categorization and deep structural formalization. The granular data demonstrates that contemporary LLMs are highly proficient at identifying linguistic phenomena when asked direct, binary questions, but they consistently fail when required to generate the underlying theoretical architectures of those phenomena (Beguš et al., 2025).

The models were most likely to succeed on subtasks requiring basic identification or feature extraction. For example, across the structural ambiguity and recursion prompts, all three models frequently succeeded on the `ident` (identification) subtasks (Beguš et al., 2025). When presented with a sentence like *"The wet rodent lab has had a leak,"* the models easily recognized the syntactic ambiguity and correctly articulated the two surface-level meanings (a lab for wet rodents versus a rodent lab that is currently wet) (Beguš et al., 2025).

Similarly, in the phonology tasks, models reliably identified the `input` and `output` phonemes of a given dataset, such as correctly noticing that an underlying oral vowel surfaces as a nasal vowel (Beguš et al., 2025). These successes suggest that LLMs are highly capable of basic pattern recognition and categorization, likely because similar descriptive explanations of language are abundant in their pretraining data (Beguš et al., 2025).

Conversely, the models were least likely to succeed when tasked with translating these surface-level observations into formal theoretical frameworks. The lowest performance across all models occurred in the tree generation and environ (phonological environment) subtasks (Beguš et al., 2025). In the syntax domain, while models could verbally explain ambiguity or movement, they struggled severely to construct accurate, hierarchical X-bar syntactic trees using LaTeX forest code. For instance, when attempting to map syntactic movement (such as *wh*-movement or subject-auxiliary inversion), models like MiniMax M2.7 and Claude Sonnet 4.6 frequently failed to properly co-index the moved constituents with their corresponding traces in the underlying tree hierarchy (Beguš et al., 2025).

This limitation was mirrored in the phonology domain. While models could extract the input and output sounds, they routinely failed to accurately formalize the environ rule using standard natural-class notation (Beguš et al., 2025). For example, when tasked with formalizing a rule where oral vowels become nasalized before nasal consonants, models struggled to abstract the segments into the proper formal notation (e.g., $V \rightarrow [+nasal] / \_[+nasal]$) (Beguš et al., 2025). This discrepancy highlights a fundamental limitation in current AI architectures: while LLMs excel at mimicking the behavioral use of language, applying rigid, abstract mathematical and linguistic formalisms to novel, uncontaminated data remains a highly complex metacognitive hurdle (Beguš et al., 2025).

6. Conclusion

The empirical results of this study demonstrate that while contemporary large language models continually claim unprecedented capabilities on traditional leaderboards, these metrics frequently misrepresent their true generalized reasoning skills. These artificially inflated scores are heavily driven by data contamination, as models inadvertently ingest static benchmark material during their pretraining phases. When subjected to the continuously updated, uncontaminated framework of LiveBench, the accuracy of all evaluated frontier models plummeted by over 50% (White et al., 2025). This severe performance degradation confirms that much of the perceived intelligence of modern AI systems stems from statistical familiarity and rote memorization rather than independent, real-world problem-solving capabilities (White et al., 2025).

Furthermore, the evaluation of expert-level domain knowledge underscores a critical limitation in current AI architectures. Within the domain of computational linguistics, the data reveal a stark contrast between surface-level pattern recognition and deep theoretical analysis (Beguš et al., 2025). While models can adequately identify general linguistic phenomena or correctly answer basic, closed-ended questions, they consistently fail when tasked with deep structural formalization, such as generating hierarchical syntactic trees or applying abstract phonological rules to novel data (Beguš et al., 2025). This indicates that while LLMs excel at mimicking behavioral language use, they still lack the underlying metacognitive frameworks necessary to actively and formally theorize about complex structures (Beguš et al., 2025).

Ultimately, these findings carry significant implications for the future of artificial intelligence development. As long as the industry relies on static, easily memorized tests like the MMLU, true cognitive progression will remain obscured by an illusion of competence (Hendrycks et al., 2021; White et al., 2025). Moving forward, the field must fundamentally shift its evaluation and training paradigms. To accurately gauge and foster genuine, generalized reasoning, developers must pivot toward dynamic, continuously updated evaluation frameworks that systematically prevent test-set leakage (White et al., 2025). Furthermore, future training methodologies must transcend broad trivia and prioritize rigorous, domain-specific metacognitive skills, ensuring that next-generation models possess the true analytical capacity required for expert-level theoretical abstraction (Beguš et al., 2025).

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